

ADTA 5560: Recurrent Neural Networks for Sequence Data

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Final Project

1. Overview

The final project covers all the topics discussed during the course. The materials in any format posted for the class activities should be considered and used for the project. Additionally, the student can use any other source of information that they can gather.

The student must create an MS Word document named “**ADTA_5560_final_project.docx**” that contains all their work, except the Python coding.

IMPORTANT NOTES:

-) *The student should present their work for each section using text and images.*
-) *The sources can be from class lectures, assignments, etc., or other sources*
-) *One picture is worth 1000 words. Adding one or more images along with the text would be much more convincing. However, an image with text explaining what it is and what it is for is considered complete.*
-) *Images can include the screenshots the student has taken while working on the classwork.*

IMPORTANT NOTES:

-) *When discussing a topic or answering a question, the student has to provide adequate explanation and supporting details to their presentation.*

IMPORTANT NOTES:

-) *If MS Docx is the document format required for submission, the student must submit the contents as MS Docx files, **not** submit PDF documents.*
-) *However, before submitting, the student should **make a backup copy of the documents** by converting them into PDF files that could be used for re-submission if the submitted file was corrupted.*

IMPORTANT NOTES:

-) *If Jupyter Notebook documents are required in a Submission Requirement, the student must **submit the Jupyter Notebook documents**, i.e., ***.ipynb documents**, **not** python programs, i.e., **not** *.py programs.*

2. PART I: A Time-series Data Set (10 Points)

TO-DO

- > Search on the Internet, using Google search or any other approach, to find a time-series dataset in the public domain, i.e., available for use without restrictions.
- > This dataset may contain either numeric values or multimedia data, including text, images, videos, and audio files.
- > The dataset should **not** be one of the datasets that have been used in lectures or classwork. It should not be made-up data such as the sine-wave data used in HW 3.
- > The dataset should **not** be one that contains the stock prices of a publicly listed company.
- > **Download** the data set.

SUBMISSION REQUIREMENT #1:

- > Write a report on the dataset that includes (but not limited to) all the critical information of the dataset, e.g., name, data type, an official website where to get the data, links to download, the time-series span (how long the time span is), the data (how many data points), the data structure of the data contained in the dataset, and so on.
- > If **possible (data: numeric or categorical)**, **submit the data set** as a **.CSV** file.

IMPORTANT NOTES:

--> *Some suggestions for public time-series data*

- *Business and/or Economic data (but not the stock price of a publicly listed company)*
- *General science and health science data*

3. PART II: RNN: Simple RNN with Sine Wave Data (10 Points)

TO-DO

Generate a sine-wave data set: The student determines all the parameters, such as the range and the number of data points, to generate the data set.

IMPORTANT NOTES:

--> *The student must generate **new sine-wave data** that is **different** from the data used in HW 3 and the midterm.*

Build, train, and evaluate a simple recurrent neural network (a complete simple RNN) with **two layers:** a SimpleRNN and a fully connected layer in Keras. The student must determine how many neurons are used in the **SimpleRNN layer**.

All the code related to the project is done **in Python** using **Jupyter Notebook**.

IMPORTANT NOTES:

--> *The student is expected to know how to count the layers of a deep neural network.*

--> First, **draw a diagram** to describe the architecture of the neural network with all the layers using MS PowerPoint or Draw Tool in MS Word.

--> **Build and train** the model using the Keras sequential model.

--> **Run all the steps** of the project in the Jupyter Notebook document to get the results of each step.

IMPORTANT NOTES:

--> *The student is expected to complete all the steps, including those related to data pre-processing.*

--> **Test** the model on the test data.

--> **Write a report** on these results.

SUBMISSION REQUIREMENT #2

--> Add **one section** to discuss the design of the neural network into the MS Word document, “**ADTA_5560_final_project.docx**,” including the **diagram** of the neural network.

--> Add **another section** to report the results of training and evaluating the model into the document: “**ADTA_5560_final_project.docx**.”

--> **Submit** all the **code** and the **results** of running the code in a **Jupyter Notebook document (*.ipynb)**

4. PART III: RNN: LSTM Neural Network (20 Points)

SUBMISSION REQUIREMENT #3

Question 3.1:

--> Explain the vanishing gradient problem

--> Explain the exploding gradient problem

Question 3.2:

--> Discuss the limitations of the SimpleRNN neural network

Question 3.3:

--> Explain how the LSTM neural network can provide powerful solutions to both gradient problems (Vanishing and Exploding) and address the limitations of the SimpleRNN neural network.

IMPORTANT NOTES:

--> *The student must include necessary figures and diagrams in their discussion.*

5. PART IV: RNN: LSTM with Time-Series Data (20 Points)

TO-DO

Collect a time-series data set with at least 500 data points:

- The data set has **only one numeric attribute** or variable.
 - The student can pre-process the original data set to achieve this.

IMPORTANT NOTES:

--> It can be the data set chosen as the solution to PART I or a different one.

--> It can be a stock price dataset, but it must **be different** from the dataset used in HW 4.

Design the network:

- The student should **decide** the **values** of **relevant parameters**:
 - **Percentage** of data for testing
 - **How many layers** of LSTM?
 - **Number of neurons** in each LSTM layer?
 - Any **DropOut** layer?
 - If with DropOut layer: the **percentage to drop**
 - **Length** of the time-series input sequence
 - **Batch size** for training (**NOTES**: *Batch size for testing and forecasting must be ONE*)
 - **Number of epochs** for training

--> **Build an LSTM neural network** using the Keras sequential model

--> **Train** the model using the training data, which is part of the collected data set.

--> **Test** the model using the testing data allocated from the collected data set.

--> **Retrain** the model using the entire collected data set and prepare it for forecasting.

--> **Perform the forecasting** using the retrained model.

--> **Run all the project steps** in a Jupyter Notebook document to get the results of each step.

IMPORTANT NOTES:

--> The student is expected to know how to count the layers of a deep neural network.

--> The student should complete all the steps, including those related to data pre-processing.

--> First, **draw a diagram** to describe the neural network's architecture with all the layers using MS PowerPoint or Draw Tool in MS Word.

--> All the **coding** is done in **Python** using **Jupyter Notebook**.

--> **Write a report** on the results.

SUBMISSION REQUIREMENT #4

--> Add **one section** to the MS Word document, "**ADTA_5560_final_project.docx**," to discuss the neural network design, **including the neural network diagram**.

- > Add **one section** with the title “**Summary of Core Parameters**” that lists all **crucial relevant parameters and their values** (number of LSTM layers, number of neurons per LSTM layer, number of DropOut layer, length of the input sequence, batch size, etc.)
- > Add **another section** into the document: “**ADTA_5560_final_project.docx,**” to report the results of all these above steps (**build, train, test, retrain, and forecast**)
- > **Submit** all the **code** and the **results** of running the code in a **Jupyter Notebook document (*.ipynb)**

6. PART V: Redesign the Neural Network (10 Points)

TO-DO

--> Using critical thinking and the experiences of working with time-series datasets, the student should **propose some changes to the above network (PART IV) or the network training process**, with which the network performance may be improved.

Redesign the network:

- The student should **decide** the **values** of **relevant parameters**:
 - **Percentage** of data for testing
 - **How many layers** of LSTM?
 - **Number of neurons** in each LSTM layer?
 - Any **DropOut** layer?
 - If with DropOut layer: the **percentage to drop**
 - **Length** of the time-series input sequence
 - **Batch size** for training (**NOTES**: *Batch size for testing and forecasting must be ONE*)
 - **Number of epochs** for training

- > **Write a proposal** to make changes to improve the performance
- > **Make the changes** to the network or the network training process as proposed in coding
- > **Build, train, and test** the **redesigned** LSTM neural network on the same dataset
- > **Retrain the redesigned** LSTM neural network and prepare it for **forecasting**
- > **Perform the forecasting** with the **redesigned** LSTM neural network
- > **Write a report** on the results

IMPORTANT NOTES:

- > *The student’s proposal to make changes to the network or the network training process **may or may not** result in real improvements in the network performance.*
- > *It is **OK** if the changes do **not** improve the performance because this is an experiment – part of the project.*
- > *The most important achievement should be that the student has used what they have learned and critical thinking to develop **reasonable** proposals to make changes.*

SUBMISSION REQUIREMENTS #5:

- > Add **one section** into the MS Word document, “[ADTA_5560_final_project.docx](#),” to discuss the new neural network design, [including the redesigned neural network diagram..](#)
- > Add **one section** with the title “**Summary of Updated Core Parameters**” that lists all **crucial relevant parameters and their values** (number of LSTM layers, number of neurons per LSTM layer, number of DropOut layer, length of the input sequence, batch size, etc.) applied to the redesigned network.
- > Add **another section** into the document: “[ADTA_5560_final_project.docx](#),” to report the results of [all these above steps \(build, train, test, retrain, and forecast\)](#).
- > **Submit** all the **code** and the **results** of running the code in a **separate Jupyter Notebook (*.ipynb)**

7. PART VI: Compare Network Performance (10 Points)

TO-DO

- > **Compare their performance** based on the design and the core parameters applied to the LSTM neural network in PART IV and the network in PART V.
- > **Write a report on** the results of comparing the performance of the two networks and to **provide a reasonable explanation** to such results:
 - Some improvement has been achieved.
 - OR no significant change in the performance has been observed.

SUBMISSION REQUIREMENTS #6:

- > Add **one new section** into the MS Word document, “[ADTA_5560_final_project.docx](#),” for the [above report](#).

8. PART VII: Project Report (20 Points)

TO-DO

- > Complete the project report ([ADTA_5560_final_project.docx](#))

SUBMISSION REQUIREMENTS #7:

The student must submit the following additional sections to the report:

- (1) Write an **introduction section** to introduce the project
- (2) Describe **what** the student **has done** in PART II, III, IV, and V in the final project
- (3) List **what** the student has **learned from the models** (Simple RNN and LSTM) and what he/she has **experienced while working with them** (build, train, test, and forecast)

- (4) Write a **conclusion section** to conclude the project report.
- **IMPORTANT NOTES:**
 - > A final project **with** the **above-mentioned additional sections** (Introduction, What has Done, Learned/Experienced, Conclusion) is considered **complete**.
 - > The student should ensure that they have included all of them in the final project.

9. PART VIII: Final Presentation Videos: YouTube Links

The student must submit the video(s) to **YouTube** as discussed in the document: “[howto_submit_videos_to_youtube.pdf](#)” posted in the module .../FINAL_PROJECT.

SUBMISSION REQUIREMENT #8:

--> Provide **all the YouTube links** of the submitted final presentation videos

10. Grading Criteria

The **final project**, including the final presentation, is **graded** based on the following **grade components**:

- | | |
|-----------------------------------|-----|
| 1. Final project report: | 70% |
| 2. Final project presentation: | 20% |
| 3. Final project peer-evaluation: | 10% |

10.1 Final Project Report (100 Points)

The **final project report**, an MS Word, contains the following **sections**:

- PART I: A time-series data set
- PART II: Simple RNN neural network with Sine Wave Data
- PART III: RNN: LSTM Cells: Core Concepts
- PART IV: LSTM neural network with Time-series Data
- PART V: Redesign the LSTM Neural Network to improve performance
- PART VI: Compare Network Performance
- PART VII: Project Report
- PART VIII: Final Presentation Videos: YouTube Links

10.2 Final Project Presentation (100 Points)

10.2.1 Overview

The student must make **a video of 30 minutes** to present their final project report to the whole class. A video **shorter than 27 minutes** is considered too short; one **longer than 33 minutes** is considered too long. Points will be deducted from either case.

The contents of the video should cover all the sections in the final project.

The final project presentation is graded based on the following grade components:

1. Contents of the presentation: 70%
 - a. i.e., How many % of the final project report is covered in the video?
2. Style of the presentation: 30%
 - a. i.e., Does the student present the topics well, clearly, and interestingly?

IMPORTANT NOTES:

--> The student can create **2 or 3 videos**, of which **the total length is about 30 minutes** instead of one video of 30 minutes if he/she chooses to do so.

--> The student **cannot** submit more than 3 videos for their final project presentation.

IMPORTANT NOTES:

--> It is strongly recommended that all the students create a separate document, e.g., PowerPoint slides, and use it to make the final project presentation in the video.

--> If the student creates a separate document, e.g., PowerPoint slides, and uses it to make the final presentation in the video, they must submit this document along with the final project report.

IMPORTANT NOTES:

--> The student can use any software tool or online service to create the video.

--> If they want to look for an online service but not sure which one provides a decent service, the student can try: screencast-o-matic.com

10.2.2 Video format

The submitted video should be an **MP4** file that is the format supported (and even the default one) by almost all the software tools and online services to create videos.

10.2.3 Naming final presentation videos

If the student submits only one video to make the final presentation, the video should be named as follows: **final_project_presentation_video_<full name>.mp4**

For example: **final_project_presentation_video_JohnSmith.mp4**

If the student **submits more than one video**, they must **name the videos in their order**.

For instance:

[final_project_presentation_video_JohnSmith_1.mp4](#)

[final_project_presentation_video_JohnSmith_2.mp4](#)

[final_project_presentation_video_JohnSmith_3.mp4](#)

10.2.4 Submitting Final Presentation Videos

The student must submit the video(s) to **YouTube** as discussed in the document: “[howto_submit_videos_to_youtube.pdf](#)” posted in the module .../FINAL_PROJECT.

10.3 Final Project Peer Evaluation (100 Points)

The instructor grade the student’s final project (the final project presentation, the project report, and all the submitted documents) and gives each student feedback.

Additionally, as a way to help each student find out how the classmates think about his/her work on the final project after watching the video, the student also receives **peer-evaluation opinions** from three **anonymous** classmates.

Each student is recommended to watch all the classmates’ videos. He/she is also required to **provide peer-evaluation opinions** on the videos of **three classmates** the instructor has randomly assigned. The evaluation will be done based on the instructor’s guidelines. It is strongly advised that the names of the **classmates** who will **receive the peer-evaluation opinions** are **strictly confidential**. The student provides peer-evaluation opinions by **filling out a form** and sending it back to the instructor by the due date.

IMPORTANT NOTES:

--> The student should make the best effort on the final project presentation video because your classmates can only access and peer-evaluate your work on the project by watching it.

11. HOWTO Submit

11.1 Final Project Report and All Related Documents

The student must submit all the required documents – Microsoft Word and Jupyter Notebook documents – as attachments to a UNT email sent to the instructor (Thuan.Nguyen@unt.edu)

The subject of the email: “**ADTA 5560: Final Project – Submission.**”

Due date & time: 8:00 AM – Monday 12/09/2024

IMPORTANT NOTES:

--) Due to the limited time for grading and posting the grades as required by the Registrar's Office, submission must be on time.

11.2 Final Project Presentation: Videos and All Related Documents

The student must submit the video to present their work on the final project to the whole class by uploading it to YouTube.

Due date & time: 8:00 AM – Monday 12/09/2024

IMPORTANT NOTES:

--) Due to the limited time for classmates to access and watch the presentation videos as required, submission must be on time.

11.3 Final Project Peer-Evaluation Forms

The student must submit their peer-evaluation forms on three classmates as attachments to a UNT email sent to the instructor (Thuan.Nguyen@unt.edu).

The subject of the email: “**ADTA 5560: Peer Evaluation Forms– Submission.**”

Due date & time: 8:00 AM – Wednesday 12/11/2024

IMPORTANT NOTES:

--) Due to the limited time for grading and posting the grades as required by the Registrar's Office submission must be on time.

IMPORTANT NOTES:

*--) The student must submit **three peer-evaluation forms** in the MS Word format, one for each evaluated classmate.*